

Predicting pitcher ERAs in a given year using conventional stats

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Executive Summary

- **Purpose**

- This project is exploring which conventional pitching statistic can predict a pitcher's ERA in a given year and develops a statistical model predicting ERA, while discussing if using conventional stats is a good fit.

- **Data and Methods**

- The data set for this project was obtained from kaggle and has data from 2015-2024.
- Predictors examined: WHIP (Walks and Hits per Inning Pitched), HR.9 (Home Runs allowed per 9 Innings Pitched), K.9 (Strikeouts per 9 innings pitched), age, team, and season fixed effects.
- A fixed effects model was used to control for season-to-season variation.
- The model performance is evaluated by R-Squared, significance testing, and through comparison

- **Key Findings**

- WHIP is the strongest predictor of ERA, while HR.9, K.9, and team all have a statistically and practically significant impact on ERA as well.
- Age was statistically significant but had very small practical significance.
- Season fixed effects captured some unobserved variation.
- Team dichotomous variables were used for 4 statistically significant teams to account for more variation in ballpark and defense differences between teams.
- The final model explains a strong share of ERA variation, with an r-Squared score of .7888.

- **Implications**

- MLB teams can predict pitching performance reliably by using a small set of conventional metrics (WHIP, HR.9, K.9).
- The results can be compared with more advanced stats to see which is a better measure of predicting pitcher performance.

- **Conclusion**

- ERA can be predicted by using a small set of conventional statistics and fixed-effects modeling. The findings can provide guidance for player evaluation and resource allocation when signing players within professional baseball.

Introduction

Baseball is a very popular sport not only in America but in many Asian and Latin American countries as well. Its impact has spanned over 100 years and influences popular culture and linguistics. As America's past time, it is undoubtedly an important facet when looking at American culture throughout the years.

As the world has moved more digital and relied on computers more, all aspects of life have changed. Sports is no different in this regard. New types of stats and way of predicting performance have come along and have impacted sports front offices to re-evaluate how they evaluate players when signing and paying them. Of the 4 most popular sport leagues that exist in the US, Baseball is one of the pioneers in these analytics used by sports front offices.

Publications on analytics in baseball have existed since 1964 and sabermetrics in baseball have existed since the 1980s. These analyses and sabermetrics provide an empirical analysis into performance of each player and teams overall. Many of these saber metrics are not the typical statistics that people have historically used when looking at player performance like earned run average or batting average. The goal of these sabermetrics is to quantify more about the sport and provide more unbiased, mathematical ways to evaluate players. There are 4 main aspects that are looked at in sabermetrics, including hitting, pitching, base running, and fielding. However, despite existing for many years, there is still push back between if sabermetrics are a good tool to be using and if old school "eye tests" and traditional statistics are still sufficient in predicting player performance.

This project will focus on the metrics of pitching and attempt to explore the conventional statistics that can be used to predict earned run average (ERA), a main statistic for evaluating a pitchers performance, used by MLB front offices and other sports analysts. It will then discuss if using traditional statistics to predict pitchers performance is still a good tool, or if potentially using more advanced sabermetrics is better. We hypothesize that WHIP will have the most impact on ERA with age, team, strikeouts per 9 innings, and home runs per 9 innings all having a significant effect on ERA as well.

Literature Review

Since the idea of using regression and statistics has existed in baseball for many years, there are many other sources that have explored a similar topics. To narrow down the statistics to look at as the explanatory variables for ERA, the paper *Predicting Earned Run Average in Major League Baseball* by Paul C. Brendel was a valuable tool to look at. This paper goes into predicting ERA of pitchers using simple linear regression.

It looks at 4 different predictors of individual metrics for a pitcher, including strikeout to walk ratio, fastball velocity, first pitch strike percentage, and groundball to flyball percentage. These statistics look

Data and Identification Strategy

Data

The data for this project was obtained from a data set on kaggle. It has pitching data from 2015-2024 in both the hitting and pitching categories and got its data from publicly available sites like ESPN and fangraphs that have MLB data. Our variable of interest is earned run average and we are looking at the pitcher-year level.

The sample obtained for the project has 3141 observations. Each observation is a pitcher for one specific year that has at least 50 innings pitched. By limiting the data sets to pitchers with at least 50 innings pitched in a year, it allows for the pitchers who have very little innings and very outlier stats to be filtered out and not skew the results.

Results

Discussion Conclusions

References

Beneventano , P., Berger, P., & Weinberg, B. (2012). Predicting Run Production and Run Prevention in Baseball: The Impact of Sabermetrics. *International Journal of Business, Humanities and Technology* , 2, 67–75.

Brendel, P. C. (n.d.). Predicting Earned Run Average in Major League Baseball .

Appendix